01. FPGA I/O + ASIC INTERFACES + 25 MHz CLOCK | Trenz p5

1. Function: U1 / J1 / J2: Route 108 active signals for eight 1.5 V ASICs through two 70–contact Hirose DF40C–70DS mezzanines, one on each long board edge, with 7 grounds each (14/15/27/34/41/50/66). J2.68, J2.69 and J2.70 carry U16–regulated VIN_3V3 to the FPGA–LDO board; 13 J1 contacts (29–33, 35–37, 39, 43, 45, 47, 49) and J2.40/J2.42 are spare no–connects. Contact order follows FPGA–LDO pin map V5 (J2 = ASIC1/3/5/7 + 14 slow nets, J1 = ASIC2/4/6/8 + 6 slow nets); each ASIC group (CLK32 + SYNC + READ + 8 data) sits in one FPGA bank with its clock on a P–side MRCC/SRCC ball.

2. Source / changes: Codex custom ASIC interface. V8 FPGA–side pin remap (Howard approved 2 Nov 2026): ASIC groups moved to the bank nearest their connector, each returned clock on a clock–capable pin. V11 (Howard approved 2 Nov 2026): J1/J2 DF40C–60DS to DF40C–70DS (Hirose DF40 catalog Jun 2024 p.8 and p.14 Note 3, guideline ETAD–H1015–00 sec 1.2.1, https://www.hirose.com/en/product/series/DF40); the V11 reading that corner contacts are not energized is superseded: Gerald (7 Nov 2026) confirmed DF40C corner contacts 1/2/69/70 are live; the fittings are separate MP pads. ASIC_REFERENCE.md sections 2–5; Chip_FPGA Interface.pptx slides 13–17/25–28; Pins.xlsx. Trenz TE0714 REV04 p5 I/O bank use adapted. https://www.trenz–electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH–TE0714–04–42I–7–L.PDF V30 (Howard 4 Nov 2026): R25 0402 –> 0201 Yageo RC0201 1 %, value unchanged.
V43 (Howard 9 Nov 2026, option (a) 'Re–pin in the FPGA', Claude): contact order = FPGA–LDO chat remap V3 (2026–11–09_FPGA_Work/03_FPGA_LDO_J1_J4_Plan_and_Remap_V3/remap_table_V3.csv; J2 = V2) so the LDO board routes J1–>J4 and J2–>J3 as ordered buses; U1 ball assignment re–solved on the 8L V1 lane plan (all 8 groups 11/11 in their clock's bank: bank 14 ASIC1 P15, ASIC8 T14; bank 15 ASIC2 D13, ASIC3 E13, ASIC4 E16; bank 34 ASIC5 R2, ASIC6 P4, ASIC7 R3; AMD xc7a35tcsg325 package file); 103 U1 nets reassigned, 14 balls freed (no–connect). BOARD2/4_CLK, BOARD2/4_SPI_CLK and NONSTIM_BOARD1/2_SPI_DL/DR removed: the PCB–5 ASIC board leaves J3/J4 pins 2/4/6/8/24/58 open, so stacked boards share the near board's clock/SPI/STIM lines. STIM_EN/START/CHB/CLK_SHARED now on J1.59/63/57/61 (FPGA–LDO 10k idle pull–downs move with them). J1.69/J1.70 added to VIN_3V3 (LDO board about 155–215 mA plus start–up vs 0.3 A per contact; all of it still comes from J3.19 VIN_5V through U16). Unverified: Gerald/Howard to confirm identical configuration of stacked boards; XDC/gateware pin constraints not yet written; ASIC timing at the new lane lengths (longest ASIC1 CLK32 J1.53 –> P15, about 12 mm); mating LDO V26 schematic.
V44 (Howard 9 Nov 2026, "Yes, study re–pin"): ASIC6_DATA6 moves from P6 to J6 (both bank 34, ASIC6 group stays 11/11 in its clock bank): its planned lane passed 0.06 mm from the JTAG TDI T9 dog–bone via. J2.59 and every other J1/J2 contact are unchanged. Codex re–solve, unverified in Vivado.
V45 (Howard 10 Nov 2026, "B: both on J2" and "Yes, T14"; Claude): edge–connector layout for the 8L V3 board (J1 east edge, J2 west edge). Contact order = FPGA–LDO chat pin map V5 (2026–11–10_FPGA_Work/15_FPGA_LDO_Pin_Map_V5_and_Schematic_V30/J1_J2_pin_map_V5.csv); this supersedes the V43 remap V3 order and the V43/V44 ball list. U1 balls re–solved by minimum total ball–to–contact distance on the 8L V3 placement: bank 34 ASIC1 N3, ASIC3 P4, ASIC5 R2, ASIC7 R3; bank 15 ASIC2 D13, ASIC4 E16; bank 14 ASIC6 R16, ASIC8 P15. CLK25_PL moves N3 –> T14 (IO_L13P_T2_MRCC_14) to free the fourth bank 34 clock ball. Nine J2 slow nets use bank 15 balls A10/B10/B11/C8/C9/D8/D9/D10/D11. VIN_3V3 moves from J1.68–70 to J2.68–70 (still three contacts). Unverified: no lane routing tried yet; ASIC timing at the new lane lengths (longest straight line 14 mm slow, 11.5 mm ASIC1); XDC/gateware constraints not written; mating FPGA–LDO schematic V30; clock placement in Vivado.
V46 (Howard 10 Nov 2026, "Yes, re–pick first" and "finish it up"; Claude): J2 FPGA balls re–picked so that every J2 lane is an ordered bus on the proposed 12–layer stack: 53 U1 balls change, the J1 balls and every J1/J2 contact are unchanged (pin map V5 frozen by Howard). Clock balls: ASIC1 R2, ASIC3 P4, ASIC5 R3, ASIC7 N3 (P–side MRCC/SRCC, bank 34; the four J2 groups stay whole in bank 34). Slow nets: BOARD3_SPI_CLK V8, STIM_CHIP3_SPI_DL/DR V7/V6, STIM_EN J6 (bank 34); STIM_CHB V9 (bank 14); STIM_CHIP1_SPI_DR/DL G14/C11, SPI_LATCH D11, BOARD1_SPI_CLK D10, FE_RESET D8, IMP_TST C8, CHIP_RESET C9, STIM_CLK B11, STIM_START B10 (bank 15). Method: integer program over ball, lane layer and channel slot (L10 for the east pad row, L8/L6 for the west pad row and J2.60/62/64), then native KiCad copper on a scratch trial copy: J2 58/58 and J1 50/50 connected ball to contact, DRC 0 errors. Unverified: the 12–layer stack and its via rules are not confirmed by JLC; eight power via stacks were moved in the trial and no planes exist yet; ASIC timing (longest ASIC lane 14.0 mm = ASIC1 clock; group spreads 4.2–6.7 mm); XDC not run in Vivado; the V45 statement about nine bank 15 slow balls is superseded by this list.

1. Function: Y2: 25MHz fabric clock drives T14 MRCC (bank 14, 1.5V). L3 filters the 1.8V VCCAUX supply; C88=100nF; R44 150R / R63 680R attenuate and damp the output.

2. Source / changes: Trenz p5 U8/L2/C24/R23 structure adapted: SiT1615 1.5V replaces SiT8008 1.8V; CLK25 used N3 SRCC until V44; V45 (Howard 10 Nov 2026, Claude): T14 MRCC, R44/R63 not moved. C88 retained100n per SiTime; R44 exact Trenz RC0201FR–0733RL restored. https://www.trenz–electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH–TE0714–04–42I–7–L.PDF https://www.sitime.com/datasheet/SiT1615 Timing/firmware constraints remain unverified.
V39 (Howard 6 Nov 2026, JLC sourcing swap): SiT1615 has no source; Y2 = YXC OT201625MJBA4SL (YSO110TR, 1.8–3.3 V, 2016, pin 1 tri–state tied high: p1 tri–state table, p2 pin map) on VCCAUX_1V8 through L3. R44 33R –> 150R and new R63 680R at the clock ball: 1.8 x 680/830 = 1.47 V high (AMD DS181 Table 2: VIN <= VCCO + 0.2 V = 1.7 V).
https://www.lcsc.com/datasheet/C669080.pdf Unverified: XO drive into 830R (datasheet specifies 15 pF only, VOH >= 90% VDD), edge rate and jitter at T14 (V45: about 8 mm of trace after the divider), LCSC numbers for R44/R63.

02. GTP CHANNELS + 125 MHz REFERENCE | Trenz p6

1. Function: J3 carries the GT recording pair (3/5, TX0 FPGA–>XEM, DC to the cable), the LVDS 32 MHz cable clock (6/8 –> R61 –> P14/R15 MRCC), LVDS command with SYNC folded in (9/11 –> R62 –> M14/N14), a separate LVDS SYNC (12/14 –> R64 –> B9/A9, V44), JTAG and 5 V (19 –> VIN_5V –> U16); U17/U18 clamp ESD on the six LVDS lines.

2. Source / changes: Codex custom HDMI map. V38 (Howard 5 Nov 2026, cable clock/SYNC/GT command), cable map confirmed with Dr. Xie: 6/8 and 9/11 move from GTP TX1/TX2 (C202–C205 removed; F2/F1/D2/D1 open per UG482 v1.9 p236) to bank 14 LVDS_25 inputs; 12/14 move from V3/V2 DIFF_SSTL15 to GTP RX0 (ground ties removed).
LVDS_25 inputs in the 1.5 V bank with DIFF_TERM=FALSE and external R61/R62 100R: AMD UG471 v1.10 p.92 and Fig.1–70; VIDIFF>=100 mV and VIN<=VCCO+0.2 V: DS181 v1.27.1 Tables 2 and 11. https://docs.amd.com/api/khub/documents/IbGcnPFe6eF19RHma_Y–IA/content
All GT AC coupling (100 nF) sits on the HDMI Extension Board: command at the XEM transmitter (C729–C744), recording TX0 at the XEM receiver (C745–C760, C200/C201 removed here); UG482 v1.9 pp.235–236 only requires it between TX and RX. https://docs.amd.com/api/khub/documents/Xk_zMbXsJj92suJ4_0Akkg/content
U17 TI TPD4E02B04 SLVSD85B (same as HDMI Extension Board U801–U808), Codex placement on the cable side. https://www.ti.com/lit/ds/symlink/tpd4e02b04.pdf No failsafe bias (10k/100R/10k gives 7.5 mV < 100 mV): firmware ignores SYNC/CMD until the MMCM locks.
Unverified: LVDS VIN margin if VCCO_1V5 < 1.45 V (XEM DS931 VOCM<=1.425 V, VOD<=454 mV), GT link rate/protocol, cable SI, shell bonding, XEM drive into an unpowered Frontend.
V40 (Howard 6 Nov 2026, HDMI map without GT command): 9/11 carry command + SYNC on the same pins and R62 (nets CABLE_CMDSYNC_P/N); 12/14 become a spare pair (CABLE_GT_SPARE_P/N), still wired to GTP RX0 E4/E3 for a later GT command; the XEM side leaves 12/14 open and fits C729–C744 only with GT. Unverified: UG482 unused–RX handling of E4/E3 on an open cable pair.
V41 (Howard 7 Nov 2026, HDMI board separate SYNC on 12/14): HDMI Extension Board V36–V41 drive a separate LVDS SYNC on 12 (P) / 14 (N) for ports 3–6 (XEM bank 64 L16/L23/L22/L18); 9/11 still carry the folded SYNC. 12/14 move from GTP RX0 E4/E3 to bank 15 IO_L1P/N D8/C8 (VCCO_1V5, LVDS_25 input, DIFF_TERM=FALSE, new R64 100R; free pair closest to J3) through new U18 TPD4E02B04 (IO1/IO2; unused IO3/IO4 tied to GND).
E4/E3 now tie to GND like RX1–RX3: AMD UG482 v1.9 Table 5–6 pp.223–224 (unused transceiver in a used Quad: MGTRXP/MGTRXN to GND). Nets CABLE_GT_SPARE_P/N –> CABLE_SYNC_P/N. Unverified: on ports 1, 2, 7 and 8 12/14 are open at the XEM (ESD only), so D8/C8 float with no failsafe bias and firmware must use the folded SYNC there; D8/C8 are in bank 15, not the cable clock bank 14, so capture uses a global clock; port–number strap or firmware selection not defined.
V42 (Howard 8 Nov 2026, 8L V1 layout): J3.3 and J3.5 swapped (J3.3 = GTP_TX0_N from U1 H1, J3.5 = GTP_TX0_P from U1 H2) so the L1 pair leaves the BGA without a crossover or length bump; gateware must set TXPOLARITY = 1 on GTP TX0 so the cable still carries the true P on 3 and N on 5: AMD UG482 v1.9 (19 Dec 2016) ch.3 'TX Polarity Control' p.106 (TXPOLARITY port), https://docs.amd.com/api/khub/documents/Xk_zMbXsJj92suJ4_0Akkg/content Unverified: TXPOLARITY = 1 in every Frontend bitstream incl. IBERT.
V44 (Howard 9 Nov 2026, "Yes, study re–pin"): 12/14 move from D8/C8 to bank 15 IO_L3P/N_T0_DQS_AD1P/N_15 B9/A9 (free since V43: the D8/C8 dog–bone sites are enclosed by the U8 / C39 / C36 / C10–link power vias, so no pair reaches them): R64 moves beside the B9 site with pad 1 on the B9 stack and pad 2 at the end of a 1.2 mm A9 fly–by stub; R22 (FLASH_CS_N pull–up) moves 0.35 mm to make room. Same bank 15 as before, so the capture note above is unchanged; 6/8 and 9/11 stay on P14/R15 and M14/N14 (a bank–15 clock–capable pair for them costs the J1/J2 lane plan: Codex V44 study). Codex choice from
erified: B9/A9 DQS/AD1 dual functions unused, LVDS SI of the fly–by stub. https://docs.amd.com/v/u/en–US/ug475_7Series_Pkg_Pinout

1. Function: U16 converts the 5 V cable feed on J3 contact 19 (VIN_5V, about 4.4–5.0 V after cable drop) into VIN_3V3, which supplies every existing regulator, logic part and J1.68. C209 local HF input, C206/C207 bulk input, C208 local output; EN tied to VIN_5V; PG unused; TP14 probes the cable voltage at the frontend.

2. Source / changes: MPS MPM3834C Rev 1.01 (6/14/2022) Fig. 4 p17 typical application; Table 1 p14 (3.3 V: R1=200k, R2=44.2k); input/output capacitor guidance p14;
VIN 2.75–6 V operating, 6.5 V abs max p3. https://www.monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/en/sku/MPM3834CGPA/
Adapted by Claude Code: R59/R60 = Table 1 values (VOUT = 0.6 V x (1 + 200k/44.2k) = 3.31 V); CFF not stuffed as Fig. 4; Fig. 4 R3/R4 EN divider replaced by direct EN tie (as U2);
input 2 x 22 uF 10 V (Fig. 4: one 22 uF); TP14 added. Unverified: cable hot–plug overshoot vs 6.5 V abs max (cable resistance should damp it; measure), real frontend current, layout and thermal.
V30 (Howard 4 Nov 2026, layout compaction): R59/R60 0402 –> 0201 Yageo RC0201/AC0201 1 %, values unchanged.
V34 (Howard 4 Nov 2026, review fixes): C209 100 nF 0201 added at the IN/EN junction with its own GND via (Fig. 3 p16: input ceramic close to IN and PGND);
R59/R60 moved directly under the FB pin; C206/C207 kept as the bulk input. MPS Rev 1.01 p16 also allows an extra 0402 input capacitor.
V39 (Howard 6 Nov 2026, JLC sourcing swap): U16 MPM3834CGPA–Z –> MPM3833CGRH–Z QFN–18 (MPM3834C unobtainable). MPS MPM3833C Rev 1.0 (12/3/2018):
pins p2, Fig.4 p17, Table 1 p15 (3.3 V: 200k/44.2k, unchanged), land p18. https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/713/MPM3833C_r1.0.pdf
SW 15, NC 11 and DNC 17/18 open. Unverified: input ripple, hot–plug overshoot (6.5 V abs max) and thermal.

1. Function: Y1 supplies125MHz LVDS through C78/C79=100nF to REFCLK1P/N. L4 filters switched3.3V; C80=470nF and C81=100nF bypass it. The REFCLK input's internal 100R is the only load; OE HIGH, pin2 NC.

2. Source / changes: Trenz TE0714–04–42I–7–L REV04 p6:125MHz LVDS U2, L1 ferrite, C10=470nF, R1=100R and C134/C135=100nF coupling. https://www.trenz–electronic.de/trenzdownloads/Trenz_Electronic/Modules_and_Module_Carriers/3x4/TE0714/REV04/Documents/SCH–TE0714–04–42I–7–L.PDF Adapted by Codex: Y1=AX3DAF1–125.0000T3 (exact Trenz oscillator MPN undisclosed), C80=470nF, C81 10uF–>100nF; added L4. External R56 removed per UG482 Fig. 5–8 p226 (100nF caps only); with it, AX3 swing is 250–450mV against DS181 350mV minimum. Abracon AX3 rev25Feb2022 pp3,9,12–13 requires active–HIGH OE and shows100nF local bypass. https://abracon.com/datasheets/AX3.pdf REFCLK1 B6/B5 matches Trenz;
P–to–P/N–to–N retained. AMD UG482 v1.9 pp225–227,235: LVDS AC–coupled reference. https://docs.amd.com/v/u/en–US/ug482_7Series_GTP_Transceivers Clock jitter, filtered–rail noise, layout and hardware startup remain unqualified. R56 uses the same100R0.1%25ppm candidate as R16; exact source MPN is undisclosed.

03. FLASH + SPI TRANSLATION | Trenz p6 adaptation

1. Function: U7/U8: 16MiB 1.8V SPI NOR flash (Winbond W25Q128JWYIQ, 21–ball WLCSP, 0.5mm pitch) and SN74AVC4T774RSV SPIx1 translation. Bank14 remains1.5V; initialCONFIGRATE3MHz/QE=1 fixed/3–byte addresses.

2. Source / changes: Package changed V26 (Howard 3 Nov 2026): WSON–8 W25Q128JWPIQ –> WLCSP–21 W25Q128JWYIQ, same W25Q128JW die, pins and Q option: QE=1 fixed, so WP#/HOLD# functions are off. W25Q128JW Rev.H 17Apr2023: WLCSP ball map Fig.1e p8, ball table 3.9 p8, WLCSP outline 10.7 p73, ordering pp75–76 (special order). https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/8919/256_W25Q128JW.pdf TI translator: https://www.ti.com/lit/ds/symlink/sn74avc4t774.pdf R19/R22 4.99k; pullups 1.8V. Codex/Claude adaptation of Trenz boot structure; no circuit value changed. Land pattern (0.25mm NSMD), body 3.0x3.0mm and via–in–pad are Claude assumptions; JLC via–in–pad, sou
V30 (Howard 4 Nov 2026, layout compaction): R19/R22 0402 –> 0201 Yageo RC0201 1 %, values unchanged. V33 (4 Nov 2026, self–review): R51–R53 47k / R54 22k bias pulls had non–existent ERA1AEB473C/223C codes; now Yageo RC0201FR 1 %, values and land unchanged.

04. CONFIGURATION + RESET OUTPUTS | Trenz p7 roles

05. FPGA SUPPLY PINS + BYPASS + XADC | Trenz p7

06. REGULATORS + READINESS SUPERVISOR | Trenz p9 roles

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Title: FPGA V54 dual60 direct FPGA/LDO interface – unrouted

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